

Positive Square Energy of Every Tetracyclic Cactus

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Abstract

We prove that every connected tetracyclic cactus G of order n satisfies $s^+(G) \geq n$. The sharp block-additive doubly nonnegative estimate reduces the theorem exactly to the cycle multisets $\{3, 3, 3, q\}$, with q odd, and $\{3, 3, 5, 5\}$. For the first family we use maximum-cycle-packing territories when $q \equiv 3 \pmod{4}$ and a shared-cut cluster territory lemma when $q \equiv 1 \pmod{4}$. For the second family, ordinary bridge cuts handle disconnected shared-cut clusters; one connected cluster has twenty exact core types. Nineteen have induced-packet certificates, while the common-root bouquet is handled by an exact two-state lobe phase theorem. All connector trees and all trees attached at core vertices are arbitrary. No cycle-closure monotonicity is assumed, and no raw, coefficientwise phase certificate is asserted where it is false.

1 Preliminaries

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are edges or cycles. For the adjacency eigenvalues $\lambda_1, \dots, \lambda_n$, write

$$s^+(G) = \sum_{\lambda_j > 0} \lambda_j^2, \quad s^-(G) = \sum_{\lambda_j < 0} \lambda_j^2, \quad D(G) = s^+(G) - s^-(G).$$

The quantities were introduced in connection with spectral coloring bounds in [1]. Liu–Tang–Zhang proved the general bound $\min\{s^+, s^-\} \geq n - 1$ [4]. Akbari–Kumar–Mohar–Pragada–Zhang conjectured the stronger conclusion $s^+(G) \geq n$ for every connected graph with at least $n + 1$ edges [3]. We prove that conclusion here for tetracyclic cacti.

A connected tetracyclic graph has $m = n + 3$, and therefore

$$s^+(G) + s^-(G) = 2m = 2n + 6, \quad s^+(G) = n + 3 + \frac{1}{2}D(G). \quad (1)$$

For every vertex partition (V_j) we use induced-subgraph superadditivity

$$s^+(G) \geq \sum_j s^+(G[V_j]), \quad (2)$$

proved in [2]. Cross edges are ignored in (2); no claim that adding an edge or closing a cycle increases s^+ is made or needed.

For positive activities $a = (a_v)$ put

$$Z_F(a) = \sum_{M \text{ matching of } F} \prod_{v \text{ unmatched by } M} a_v, \quad Z_F(t) = Z_F((t)_v),$$

and normalize the characteristic polynomial by

$$\Psi_G(t) = i^{-n} \det(itI - A(G)) = \prod_j (t + i\lambda_j).$$

Sachs' expansion, grouped by collections $\mathcal{C}$ of pairwise vertex-disjoint cycles, is

$$\Psi_G(t) = \sum_{\mathcal{C}} \prod_{C \in \mathcal{C}} (-2i^{-|C|}) Z_{G-V(C)}(t). \quad (3)$$

Thus cycles of lengths 1 and 3 modulo four have multipliers $2i$ and $-2i$, respectively. If $\Theta_G(t)$ is the continuous argument of $\Psi_G(t)$ tending to zero as $t \rightarrow \infty$, then

$$D(G) = -\frac{4}{\pi} \int_0^\infty t \Theta_G(t) dt. \quad (4)$$

This follows by integrating $\lambda|\lambda| = (2/\pi) \int_0^\infty \lambda^3/(\lambda^2 + t^2) dt$, summing, using $\sum_j \lambda_j = 0$, and integrating by parts.

Arbitrary attached trees can be eliminated exactly. Orient a rooted branch toward its core attachment. If $T_{u \rightarrow p}$ is the subtree below u , then

$$Q_{u \rightarrow p} = \frac{Z_{T_{u \rightarrow p}}(t)}{Z_{T_{u \rightarrow p} - u}(t)} = t + \sum_{w \text{ child of } u} Q_{w \rightarrow u}^{-1} \geq t. \quad (5)$$

After elimination, every retained core vertex has an activity $a_v = t + \sum Q_{u \rightarrow v}^{-1} \geq t$, and every Sachs term has the same positive factor. Splitting a matching according as its root is unmatched or matched to one child proves (5), including the terms in which a selected Sachs cycle deletes the attachment vertex.

We use the following lower-rank results. They are proved for arbitrary connectors and arbitrary attached trees in the cited companion manuscripts; the odd-unicyclic sign theorem originates with Ning–Zeng [5]. Put

$$\delta_q = \sec(\pi/q) - 1 \quad (q \equiv 1 \pmod{4}), \quad \delta_5 = \sqrt{5} - 2.$$

Lemma 1.1 (Packet bounds). *Let H be a connected cactus of order h .*

1. *If every cycle is $3 \pmod{4}$ and the cycle-packing number is at most two, then $D(H) > 0$. In particular, a packet with $r = 1$ or 2 triangles has $s^+(H) > h + r - 1$.*
2. *A unicyclic C_q packet, $q \equiv 1 \pmod{4}$, has $s^+(H) \geq h - \delta_q$.*
3. *A C_3, C_q bicyclic packet, $q \equiv 1 \pmod{4}$, has $s^+(H) > h + 1 - \delta_q$.*
4. *Every bicyclic cactus has $s^+(H) \geq h$.*
5. *A C_3, C_3, C_q tricyclic packet with q odd has $s^+(H) > h$. If the two triangles share a cut vertex and $q = 5$, the shared-triangle phase calculation supplies the stronger bound $s^+(H) > h + 2 - \delta_5$.*
6. *A C_3, C_5, C_5 tricyclic packet has $s^+(H) > h$.*

Every assertion remains valid with arbitrary trees attached anywhere.

Proof. For (1), (3) has strictly negative imaginary part: singleton cycle terms are negative imaginary, and, because at most two cycles can be selected, all remaining terms are real. Equations (4)–(1), in the appropriate cyclomatic rank, give the claim. Tree elimination gives a unicyclic C_q packet the phase of $Z_{C_q}(a) + 2i$. Since $a_v \geq t$, its phase is at most that of the bare cycle, for which

$$D(C_q) = -2\delta_q, \quad s^+(C_q) = q + 1 - \sec(\pi/q).$$

This proves (2). The mixed C_3, C_q product-subpartition comparison proves (3), while (4) is the complete bicyclic-cactus theorem [7]. The complete tricyclic-cactus theorem [8] gives (5)–(6); its shared-triangle $\{3, 3, 5\}$ phase calculation gives the displayed stronger bound under the stated incidence hypothesis. These arguments use actual induced packets or exact matching phases, never cycle-closure monotonicity. $\square$

For later bookkeeping, Table 1 gives the credits $s^+(H) - h$ used in the finite certificates. A word “strict” means that equality with the displayed lower bound is excluded.

cycle content	credit	status
tree	−1	weak
3	0	strict
3, 3	1	strict
5	$2 - \sqrt{5} = -\delta_5$	weak
3, 5	$3 - \sqrt{5} = 1 - \delta_5$	strict
5, 5	0	weak
3, 3, 5 with shared triangles	$4 - \sqrt{5} = 2 - \delta_5$	strict
3, 5, 5	0	strict

Table 1: Proved packet credits, all allowing arbitrary attached trees.

2 The exact DNN residual classification

For a connected graph with an edge define the Liu–Tang–Zhang functional

$$\kappa(G) = \sup_{\substack{M \succeq 0, M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

Lemma 2.1 (Sharp cactus DNN bound). *Suppose a cactus has b bridge blocks. If its cyclic blocks are $C_{\ell_1}, \dots, C_{\ell_r}$, then*

$$\kappa(G) = b + \sum_{j=1}^r \kappa(C_{\ell_j}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even}, \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd}, \end{cases}$$

and $s^-(G) \leq \kappa(G)$.

Proof. We recall the sharp argument from [6]. Folding a nonedge entry M_{uv} into the diagonal by adding $M_{uv}(e_u - e_v)(e_u - e_v)^T$ preserves the denominator and edge entries. On a cycle, rotational averaging and concavity of the square root reduce the problem to common edge entry x and row sum r . A Fourier eigenvalue gives $r \geq 4x$ for even ℓ and $r \geq 2(1 + \cos(\pi/\ell))x$ for odd ℓ . Equality is attained by $x(2cI + A(C_\ell))$, with $c = 1$ or $c = \cos(\pi/\ell)$.

At a one-vertex sum, fold first and use Gram vectors. The spans on opposite sides are orthogonal. Splitting the cut vector into its two span projections splits both the edge numerator and $\mathbf{1}^T M \mathbf{1}$; Cauchy–Schwarz gives subadditivity. Orthogonal gluing of near-optimizers, scaled so $T_1/\kappa_1 = T_2/\kappa_2$, gives the reverse inequality. Hence κ is block additive and $\kappa(K_2) = 1$.

Finally write $A = P - B$ for the positive and negative spectral parts. The Schur product theorem makes $B \circ B$ doubly nonnegative, while $s^-/2 = -\sum_{uv \in E} B_{uv} \leq \sum_{uv \in E} |B_{uv}|$ and $\mathbf{1}^T (B \circ B) \mathbf{1} = s^-$. The definition of κ gives $(s^-)^2 \leq \kappa(G)s^-$. $\square$

Put $\epsilon_\ell = 0$ for even ℓ and

$$\epsilon_\ell = \ell \tan^2\left(\frac{\pi}{2\ell}\right) \quad (\ell \text{ odd}).$$

Then $\kappa(C_\ell) = \ell + \epsilon_\ell$,

$$\epsilon_3 = 1, \quad \epsilon_5 = 5 - 2\sqrt{5}, \quad \epsilon_5 + \epsilon_7 < 1, \quad (6)$$

and ϵ_ℓ strictly decreases through odd integers. Indeed, after $u = \pi/(2x)$ the monotonicity is that of $\tan^2 u/u$, whose derivative is positive because $2u > \sin u \cos u$. Also $\cos(\pi/7) > 7/8$ gives $\epsilon_7 < 7/15 < 2\sqrt{5} - 4 = 1 - \epsilon_5$.

For four cycle lengths $\ell_1, \dots, \ell_4$, block counting gives $b + \sum_j \ell_j = n + 3$. Lemma 2.1 and the energy identity yield

$$s^+(G) \geq n + 3 - \sum_{j=1}^4 \epsilon_{\ell_j}. \quad (7)$$

Proposition 2.2 (Exact residual list). *The sharp DNN estimate (7) proves $s^+(G) \geq n$ unless the cycle multiset is*

$$\{3, 3, 3, q\} \quad (q \geq 3 \text{ odd}), \quad \text{or} \quad \{3, 3, 5, 5\}. \quad (8)$$

These and only these multisets have $\sum_j \epsilon_{\ell_j} > 3$.

Proof. Four triangles give the first family with $q = 3$. With exactly three triangles, the excess sum exceeds three precisely when the fourth length is odd, giving all remaining $\{3, 3, 3, q\}$. With exactly two triangles, the two other lengths must be odd and their excesses must sum to more than one. Equation (6) and strict monotonicity leave exactly 5, 5. With at most one triangle, the sum is at most $1 + 3\epsilon_5 < 3$. Even lengths contribute zero. This exhausts all unordered length multisets. $\square$

The list is a classification of what the sharp DNN relaxation leaves, not a list of counterexamples and not a claim that the spectral inequality in the DNN proof is sharp.

3 Three triangles and one odd cycle

We first record the territory statements needed here. A cyclic territory is an induced connected vertex set containing at least one cyclic block. Every component outside a chosen cyclic core has a unique core root: two roots and a path through the component would create another cycle. Assigning each whole rooted component to the territory containing its root extends every core partition over arbitrary trees.

Lemma 3.1 (Maximum-packing territories). *In a cactus, let $\mathcal{P}$ be a maximum collection of pairwise vertex-disjoint cyclic blocks. There is a partition into connected induced territories, one containing each member of $\mathcal{P}$, such that every territory has cycle-packing number at most two.*

Proof. Use the block-cut tree, subdividing bridge-only paths as necessary. Delete edges of this tree so that each resulting component contains exactly one selected cycle node; assign a tie component consistently toward either neighbor. Lifting the components gives connected induced territories, and hanging branches follow their unique attachment. If one territory contained three pairwise disjoint cyclic blocks, none could lie outside that territory and each is disjoint from the selected cycles in all other territories. Replacing its one selected block by those three would enlarge $\mathcal{P}$, a contradiction. Thus its packing number is at most two. Notice that this is an induced partition argument, not the assertion that closing any of the cut cycles is spectrally monotone. $\square$

For the opposite residue we need a small block-tree fact. Join two cyclic blocks when they share a cut vertex; components of this relation are called shared-cut clusters.

Lemma 3.2 (The C_{333q} cluster territories). *Let a cactus have cyclic blocks C_3, C_3, C_3, C_q , where $q \equiv 1 \pmod{4}$, and let $\delta = \sec(\pi/q) - 1 < 1$. Its vertices admit induced connected packet territories whose total proved credit is positive or is a strict lower bound of zero. Every packet used has cycle-packing number at most two, is a C_q, C_3, C_q , or C_3, C_3, C_q packet, or contains only triangles; tree packets occur only in the one-attachment connected case.*

Proof. First suppose that the shared-cut graph is disconnected. Contract every shared-cut cluster and every maximal bridge-only connector; the resulting cluster incidence is a tree. Let t be the number of triangles in the cluster containing C_q . Cutting connector edges partitions this tree into induced connected cluster territories, with each intervening connector put wholly on one of its two sides. If $t = 1$ or 2 , use the rank-two or rank-three C_q territory and the remaining triangular clusters. Its credit is respectively greater than $1 - \delta$ or greater than 0 , while each remaining triangular cluster is put in a strict triangular packet (one per cluster). If $t = 0$ and some other cluster contains at least two triangles, that cluster has credit greater than at least 1 , which pays the C_q singleton deficit $-\delta$; a three-triangle cluster still has packing number at most two, because its shared-cut graph is connected and hence two triangles meet. All other triangular clusters have nonnegative strict credit. If $t = 0$ and all three triangular clusters are singletons, take the unique path in the cluster tree from the C_q cluster to a nearest triangular cluster, put that entire connector into their C_3, C_q bicyclic packet, and leave the other two triangles in strict triangular packets. Its credit is greater than $1 - \delta > 0$.

Now suppose that all four cyclic blocks form one shared-cut cluster. Remove the C_q block node from its block-cut tree. Group together the triangle blocks in each remaining component: triangles attached indirectly through other triangles therefore belong to the same group. The nonempty groups attach to distinct vertices $v_1, \dots, v_d$ of C_q ; write r_i for their triangle counts, so $r_i \geq 1$ and $\sum_i r_i = 3$.

Assume first that $d \geq 2$. Fix a cyclic orientation of C_q . Removing the attachment vertices leaves exactly d (possibly empty) open path components. Assign every internal vertex of each such path, together with every branch rooted at an internal vertex, to the group at its preceding endpoint. Thus the packet H_i consists of the i th triangle component group, v_i , and the interiors assigned to v_i . The first edge of every nonempty assigned arc joins its interior to v_i ; an empty arc requires no vertices. The edge at the other end goes to the next packet. Consequently the H_i are connected induced subgraphs and partition all vertices. No packet contains all of C_q , and the complete cycles of H_i are exactly its r_i triangles. Since $d \geq 2$ and the r_i are positive with sum three, each $r_i \leq 2$, so Lemma 1.1(1) gives packet credit strictly greater than $r_i - 1$. The total credit is therefore strictly greater than $\sum_i (r_i - 1) = 3 - d \geq 0$.

If $d = 1$, put the entire triangle component group, including its common attachment vertex v_1 , in one packet, and put the path $C_q - v_1$ in a tree packet. The triangle group is connected

in the shared-cut graph, so two of its three triangles meet and its cycle-packing number is at most two. The first packet therefore has credit greater than 2 by Lemma 1.1(1), while the tree packet has credit at least -1 ; their total credit is greater than 1. This includes the common-root bouquet. In every case each off-core component is assigned whole to the packet containing its unique root, including components rooted on assigned arc interiors, so arbitrary attached trees preserve connectivity, inducedness, and cycle content.

This is purely block-tree bookkeeping. In particular, the proof does not use the tempting raw comparison $\Phi = 2R - Z_q I$: at $q = 5$ that polynomial has negative coefficients for eight of the eleven shared core types, even after substituting $a_v = t + y_v$. The exact failed-certificate audit is listed in Appendix A; no false coefficientwise Φ claim enters this lemma. $\square$

Proposition 3.3. *If the cyclic blocks of G are C_3, C_3, C_3, C_q with q odd, then $s^+(G) > n$.*

Proof. If $q \equiv 3 \pmod{4}$, apply Lemma 3.1. Every territory has only $3 \pmod{4}$ cycles and packing number at most two, so Lemma 1.1(1) gives $D > 0$ on each. A territory with $r \geq 1$ cycles and order h consequently has $s^+ > h - 1 + r \geq h$. Superadditivity proves $s^+(G) > n$.

Let $q \equiv 1 \pmod{4}$. Lemma 3.2 and Table 1 give positive total credit, or zero total credit with at least one strict packet; either conclusion and (2) give $s^+(G) > n$. The common-root theorem below independently gives the stronger bound $s^+(G) \geq n + 3 - \delta_q$ for the bouquet. $\square$

4 Two triangles and two pentagons

4.1 Disconnected shared-cut clusters

Assume first that the shared-cut graph has more than one component. Its contracted cluster incidence is a tree. Deleting a connector bridge gives vertex sets of connected induced subgraphs. No connector length is bounded: each bridge-tree component stays wholly on one side of the cut.

Proposition 4.1. *If the C_3, C_3, C_5, C_5 blocks do not form one shared-cut cluster, then $s^+(G) > n$.*

Proof. Write $\delta = \delta_5 = \sqrt{5} - 2$. The distributions of four cycle blocks among clusters are as follows.

For a $3+1$ distribution, isolate the singleton leaf cluster. A singleton triangle is strict with zero credit. If it is a pentagon, retain the hostile partition exactly as

$$\{3, 3, 5\} \mid \{5\}. \quad (9)$$

Let H be the internal $\{3, 3, 5\}$ cluster and let the other pentagon be joined to it by an arbitrary connector tree. If the two triangles of H meet, the shared-triangle phase bound in Lemma 1.1(5) gives credit greater than $2 - \delta$ on H , while the external pentagon has credit at least $-\delta$. Thus the total credit is greater than $2 - 2\delta = 6 - 2\sqrt{5} > 0$.

Suppose instead that the two triangles T_1, T_2 are disjoint. Because H is one shared-cut cluster, its pentagon P meets both triangles, at distinct vertices x_1, x_2 . Partition its core into

$$U_1 = V(T_1), \quad U_2 = V(T_2) \cup (V(P) \setminus \{x_1\}).$$

These are induced and connected: $P - x_1$ is a path containing x_2 . Whichever packet owns the internal root of the connector absorbs the full connector and the external C_5 . On P , the root is owned by U_1 exactly when it is x_1 , and otherwise by U_2 ; a root on either triangle is owned by the packet containing that triangle. Hence the enlarged packet has cycle content C_3, C_5 and credit greater than $1 - \delta$, while the other triangle packet has strict credit 0. The total is greater than

$1 - \delta = 3 - \sqrt{5} > 0$, and hence $s^+(G) > n + 3 - \sqrt{5}$. Every branch at the connector root or at an internal connector vertex is assigned with that whole connector side, so no restriction on connector length or rooted branches is used. This is the hostile exact construction; a uniform strong bound for arbitrary $\{3, 3, 5\}$ incidence is neither asserted nor needed.

For $2+2$, the cut is either $\{3, 3\} \mid \{5, 5\}$, with total credit strictly greater than 1, or $\{3, 5\} \mid \{3, 5\}$, with total credit $2(1 - \delta) = 6 - 2\sqrt{5} > 0$.

For $2 + 1 + 1$, cut the two appropriate connector edges. According to the two-cycle cluster, the credit sums are

$$\begin{aligned} \{3, 3\} \mid \{5\} \mid \{5\} : \quad 1 - 2\delta = 5 - 2\sqrt{5} > 0, \\ \{3, 5\} \mid \{3\} \mid \{5\} : \quad 1 - 2\delta > 0, \quad \{5, 5\} \mid \{3\} \mid \{3\} : \quad 0 \text{ strict.} \end{aligned}$$

For $1 + 1 + 1 + 1$, isolate a triangular leaf if one exists. Otherwise the two triangles are internal and a pentagon is a leaf; deleting that leaf leaves a connected $\{3, 3, 5\}$ territory, so the two structural cases following (9) apply. These four integer distributions exhaust every disconnected cluster tree. $\square$

4.2 One shared-cut cluster: nineteen packet certificates

The bare cyclic core has 13 vertices and 16 edges. Recursively adjoining leaf cycle blocks, then quotienting independently by exact graph isomorphism and by canonical colored-cycle coding, gives exactly twenty types. Label the cycles T_0, T_1, P_2, P_3 . An incidence string such as $012 + 23$ records the sets of cycles meeting at each shared cut vertex.

For each core, the exact audit tests every nonempty vertex subset. It retains a subset only when the induced graph is connected, its complete-cycle content is a row of Table 1, and its cyclomatic number equals the number of retained listed cycles. A subset dynamic program anchors each new part at the least remaining vertex, thereby exhausting every set partition without packet-order duplicates. Credits are pairs $a + b\sqrt{5}$ and are compared by integer sign and squaring tests, never floating point.

Table 2 is the complete certificate ledger. Vertex labels refer to the canonical cycles printed by the script in Appendix A. Semicolons separate induced packets; every unlisted packet vertex is absent only from the abbreviated cycle-content column, not from the explicit vertex sets.

Since $5 - 2\sqrt{5} > 0$, $3 - \sqrt{5} > 0$, and type 5 has zero credit with a strict triangular packet, (2) proves $s^+(G) > n$ for types 2–20. Every off-core component has a unique core root and is assigned whole to the packet containing that root. Thus the finite core certificates apply to arbitrary attached trees; they are not certificates merely for bare cores.

4.3 Type 1: the common-root bouquet

Type 1 has incidence 0123: all four cycles meet at one cut vertex o . Its best partition using only Table 1 has credit -1 , so there is no induced-packet proof of this type. We give the exact analytic replacement.

Theorem 4.2 (Common-root bouquet phase). *Suppose r cyclic blocks $C_{\ell_1}, \dots, C_{\ell_r}$ share one common cut vertex, no ℓ_j is divisible by four, and arbitrary trees are attached at all core vertices. Then*

$$s^+(G) \geq n + r - 1 - \sum_{\ell_j \equiv 1 \pmod{4}} (\sec(\pi/\ell_j) - 1). \quad (10)$$

Proof. Eliminate all trees by (5). On lobe j , list the nonroot activities as $x_{j,1}, \dots, x_{j,\ell_j-1}$ and define the continuant

$$K() = 1, \quad K(x_1) = x_1, \quad K(x_1, \dots, x_k) = x_k K(x_1, \dots, x_{k-1}) + K(x_1, \dots, x_{k-2}).$$

The exact root-unavailable and root-matched states are

$$A_j = K(x_{j,1}, \dots, x_{j,\ell_j-1}),$$

$$B_j = K(x_{j,2}, \dots, x_{j,\ell_j-1}) + K(x_{j,1}, \dots, x_{j,\ell_j-2}).$$

If a is the root activity, the matching carrier is

$$\left(\prod_j A_j \right) \left(a + \sum_j B_j/A_j \right).$$

At most one whole lobe can be selected in a Sachs subgraph. Dividing by the common positive factor $K_{\text{tree}} \prod_j A_j$ therefore gives $X + iY$, where

$$X = a + \sum_j B_j/A_j + 2 \sum_{\ell_j \equiv 2 \pmod{4}} A_j^{-1} > 0,$$

$$Y = 2 \sum_{\ell_j \equiv 1 \pmod{4}} A_j^{-1} - 2 \sum_{\ell_j \equiv 3 \pmod{4}} A_j^{-1}.$$

For an odd lobe set $c_j = aA_j + B_j$. Since $X \geq a + B_j/A_j$, positivity and subadditivity of $\arctan$ give

$$- \sum_{\ell_j \equiv 3 \pmod{4}} \arctan \frac{2}{c_j} \leq \Theta_G(t) \leq \sum_{\ell_j \equiv 1 \pmod{4}} \arctan \frac{2}{c_j}.$$

All effective activities are at least t , so $c_j \geq Z_{C_{\ell_j}}(t)$. Hence the upper phase is bounded by the sum of the corresponding bare 1 (mod 4) cycle phases. Applying (4), using $D(C_\ell) = -2(\sec(\pi/\ell) - 1)$ for $\ell \equiv 1 \pmod{4}$, and using $m = n - 1 + r$ proves (10). $\square$

For C_3, C_3, C_5, C_5 , only the two pentagons are adverse, and

$$s^+(G) \geq n + 3 - 2(\sqrt{5} - 2) = n + 7 - 2\sqrt{5} > n. \quad (11)$$

This proves type 1. It also gives the stronger C_{333q} common-root bound quoted after Proposition 3.3. The positivity of X explains why no cycle of length 0 (mod 4) is included in this theorem: such a lobe contributes $-2/A_j$ to X .

Proposition 4.3. *Every connected cactus with cyclic blocks C_3, C_3, C_5, C_5 satisfies $s^+(G) > n$.*

Proof. If the shared-cut graph is disconnected, apply Proposition 4.1. Otherwise the exact census gives one of twenty types. Table 2 proves types 2–20, and (11) proves type 1. These alternatives include every connector and every attached tree. $\square$

5 Exhaustive conclusion

Theorem 5.1. *Every connected tetracyclic cactus G on n vertices satisfies*

$$\boxed{s^+(G) \geq n}.$$

Proof. Proposition 2.2 shows that the sharp DNN bound proves the theorem for every cycle-length multiset except $\{3, 3, 3, q\}$ with q odd and $\{3, 3, 5, 5\}$. The first complete family is settled by Proposition 3.3; the second is settled by Proposition 4.3. The residual proofs exhaust the shared-cut cluster tree, all common-root and non-common-root incidences, the exact twenty connected C_{335} cores, arbitrary bridge paths, and arbitrary rooted trees. Thus no tetracyclic cactus remains. $\square$

Remark 5.2 (What is not used). The proof never assumes that adding an edge, closing a path into a cycle, or replacing a tree by a star increases s^+ . It uses only induced-subgraph superadditivity, exact packet theorems, and exact matching-phase identities. It also does not use a raw coefficientwise phase certificate for all shared C_{33q} cores: that proposed certificate is false already at $q = 5$.

A Reproduction and proof objects

All paths are relative to the repository root. Reproduce the twenty-core C_{335} census and the exhaustive induced-packet dynamic program with

```
python positive-square-energy/experiments/
    c3_c3_c5_c5_induced_packet_partitions.py
```

The expected final line is

```
SUMMARY exact_positive=18/20 strict_target=19/20
minimum_score=-1 minimum_decimal=-1.000000000000
```

Here 18/20 counts literally positive credits; 19/20 also counts type 5, whose zero credit is strict. The missing type is exactly the common-root bouquet proved by Theorem 4.2. The script imports

```
positive-square-energy/experiments/c3_c3_c5_c5_shared_cluster_certificate.py
```

which independently generates the twenty canonical core types. The audit description and full ledger are

```
research/c3-c3-c5-c5-induced-packet-partition-audit-2026-07-26.md
research/c3-c3-c5-c5-tetracyclic-proof-2026-07-26.md
```

The failed raw C_{335} coefficient certificate is independently audited by

```
python positive-square-energy/experiments/
    c3_c3_c3_c5_shared_cluster_certificate.py
```

It finds exactly eleven shared core types and only three coefficientwise positive instances of $\Phi = 2R - Z_5 I$. The aggregate exact-ledger SHA-256 is

```
f7f39c81aa13443259ca5a644acfeb639041bc44ff393145a3a031520bac4c03
```

This negative audit is included to make the logical boundary reproducible: Lemma 3.2 is a territory-credit proof, not a disguised raw phase assertion.

The analytic common-root calculation is recorded separately at

```
research/common-root-cycle-bouquet-phase-2026-07-26.md
```

The lower-rank packet proofs and sharp DNN proof are

```
sharp-cactus-dnn/paper.tex
all-bicyclic-cacti/paper.tex
all-tricyclic-cacti/paper.tex
research/c3-c3-cq-tricyclic-cactus-2026-07-25.md
```

References

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type	incidence	packet vertex sets	cycle contents	credit
2	012+03	$\{1, 9, 10, 11, 12\}; \{0, 2, 3, 4, 5, 6, 7, 8\}$	$P_3; T_1 P_2$	$5 - 2\sqrt{5}$, strict
3	012+23	$\{5, 9, 10, 11, 12\}; \{0, 1, 2, 3, 4, 6, 7, 8\}$	$P_3; T_0 T_1$	$3 - \sqrt{5}$, strict
4	012+23	$\{6, 9, 10, 11, 12\}; \{0, 1, 2, 3, 4, 5, 7, 8\}$	$P_3; T_0 T_1$	$3 - \sqrt{5}$, strict
5	023+01	$\{1, 5, 6, 7, 8, 9, 10, 11, 12\}; \{0, 2, 3, 4\}$	$P_2 P_3; T_1$	0, strict
6	01+02+03	$\{1, 5, 6, 7, 8\}; \{0, 2, 3, 4, 9, 10, 11, 12\}$	$P_2; T_1 P_3$	$5 - 2\sqrt{5}$, strict
7	01+02+13	$\{1, 5, 6, 7, 8\}; \{0, 2, 3, 4, 9, 10, 11, 12\}$	$P_2; T_1 P_3$	$5 - 2\sqrt{5}$, strict
8	01+02+23	$\{5, 9, 10, 11, 12\}; \{0, 1, 2, 3, 4, 6, 7, 8\}$	$P_3; T_0 T_1$	$3 - \sqrt{5}$, strict
9	01+02+23	$\{6, 9, 10, 11, 12\}; \{0, 1, 2, 3, 4, 5, 7, 8\}$	$P_3; T_0 T_1$	$3 - \sqrt{5}$, strict
10	023+12	$\{3, 4, 5\}; \{0, 1, 2, 6, 7, 8, 9, 10, 11, 12\}$	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
11	02+03+12	same as type 10	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
12	02+12+23	same as type 10	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
13	02+12+23	same as type 10	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
14	023+12	same as type 10	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
15	02+03+12	same as type 10	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
16	02+12+23	same as type 10	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
17	02+12+23	same as type 10	$T_1; T_0 P_3$	$3 - \sqrt{5}$, strict
18	02+13+23	$\{3, 4, 5\}; \{0, 1, 2, 6, 7, 8, 9, 10, 11, 12\}$	$T_1; T_0 P_2$	$3 - \sqrt{5}$, strict
19	02+13+23	same as type 18	$T_1; T_0 P_2$	$3 - \sqrt{5}$, strict
20	02+13+23	same as type 18	$T_1; T_0 P_2$	$3 - \sqrt{5}$, strict

Table 2: Exact induced-packet certificates for connected shared-cut types 2–20. The script prints the canonical cycles and independently checks all vertex subsets.